Pratik Pujari

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Education

University of Colorado Boulder — Boulder, CO
Master of Science in Computer Science; GPA: 3.8/4.0 | Expected Graduation: May 2027 — *Aug. 2025 – Present*

Sardar Patel Institute of Technology — Mumbai, India
Bachelor of Technology in Computer Engineering; GPA: 8.6/10.0 — *Oct. 2020 – May 2024*

Experience

Software Engineer Intern — May 2026 – Aug 2026
Visa Inc. — *Highlands Ranch, CO*

- Architected **VisionX**, an AI log-triage platform replacing manual log triage: **Drain3** templating, embedding-based **ML** classification with **LLM** escalation, and two-pass **HDBSCAN/UMAP** clustering into **P0–P3** prioritized issues.
- Built an autonomous **remediation agent** (**LangGraph** ReAct via GitHub Copilot SDK), auto-dispatched on P0 issues, that diagnoses root cause in the failing repo and opens **GitHub PRs** and **Jira** comments via Visa's **MCP** Hub.
- Engineered a fault-tolerant **FastAPI** + **Kafka** pipeline (9 topics, retry backoff, dead-letter queue) over **PostgreSQL**, classifying each log template once and caching embeddings in **Redis/SQLite** to keep the live ingestion path fast.
- Built a predictive **risk engine** scoring per-source outage risk (0–100) from seasonal baselines, trend, and novelty.
- Prevented false alerts by excluding backfilled (600s+ lag) logs from live **Prometheus** and **Alertmanager** rules.
- Added classifier drift detection with guarded auto-retraining; demoed VisionX live to Visa's **VP-level** leadership.

Software Developer Intern — Nov. 2023 – Dec. 2024
Aim4U Software Solutions Private Limited — *Mumbai, India*

- Architected **Dermalens**, an AI telehealth platform (doctor discovery, booking, payments, e-prescriptions) on **Flutter** and **AWS Lambda**, with **Detectron2** skin-disease classification at **90–95% accuracy**; reached **300+ downloads**.
- Led the **Agora SDK** integration for HIPAA-compliant video consultations with adaptive bitrate, encrypted channels, and automatic session recovery when mobile networks fail, completing Dermalens' end-to-end telemedicine workflow.
- Co-developed **Autobuddys**, a **Flutter** app with 5 gamified modules and **Firebase** analytics; **250+ downloads**.
- Built **CI/CD** pipelines with pre-merge checks and integration tests; mentored **3 junior interns** on code review.
- Pitched Dermalens to investors at 3+ startup competitions; shipping the judges' feedback lifted downloads by **23%**.

Research and Software Intern — Jan. 2023 – June 2023
Acuradyne Systems Private Limited, IIT Bombay Research Park — *Mumbai, India*

- Re-engineered the data link of an IoT radial-pulse device for aging assessment, migrating Bluetooth Classic to **BLE** on **ESP32** with chunked reassembly for **566%** faster transfer (60 to 400 points/s; 850ms to 127ms per data payload).
- Built a **Flutter** patient app with **60fps** real-time pulse charts and **PDF** clinical reports across **50+** medical devices.
- Shipped **OTA** firmware updates over **BLE** with checksum-verified transfer and rollback, cutting production time **40%**.
- Secured **ESP32**-to-server links with **AES-128** encryption and 4-pin code authentication; led a team of **3–4 interns**.

Projects

Echo – Real-time Chat Platform | *Go, Kotlin, PostgreSQL, Kafka* | Backend | Android — Mar. 2025 – Sept. 2025

- Built a **9-service Go** backend on **PostgreSQL**, **Redis**, and event-sourced **Kafka** for message delivery guarantees.
- Built a **WebSocket** engine for **10K+** concurrent connections at **sub-10ms** latency with heartbeats and read receipts.
- Built the **Kotlin/Jetpack Compose** client as **11 Gradle modules** with MVI, Hilt, Room, and offline-first sync.
- Designed a **Compose** design system of **60+** reusable components and enforced conventions with **4** custom Lint rules.
- Added **TimescaleDB** for message-history analytics and synchronized sessions across devices via event-driven updates.

Nesters – Student Housing | *Flutter, Supabase, Firebase, GCP* | Play Store | App Store — May 2024 – Jan. 2025

- Shipped a **Flutter** student-housing app to **Google Play** and **App Store**, owning schema design to store release.
- Built real-time **Firebase** chat with **FCM** push notifications and read receipts designed for **50K** concurrent users.
- Designed roommate matching on lifestyle, schedule, and budget, plus a **Supabase** buy, sell, and rent marketplace.

JetScan – Document Scanner | *Kotlin, Compose, OpenCV, GCP* | GitHub | Demo — May 2024 – Nov. 2024

- Built an **80K+ line** open-source Android scanner in **Jetpack Compose** with **CameraX** and an **OpenCV** pipeline.
- Engineered **100ms** angle-invariant document detection using LAB color space, **Canny**, **HoughLinesP**, and contours.
- Integrated **Google Document AI** OCR and a **Flask** PDF service on **Cloud Run** for split, merge, and encrypt.

Technical Skills

Languages: Python, Go, Java, Kotlin, TypeScript, C++, SQL
Backend & Systems: Distributed Systems, Microservices, REST APIs, Kafka, FastAPI, WebSocket
Data & Cloud: PostgreSQL, Redis, MongoDB, AWS, GCP, Docker, Kubernetes
Practices & Tools: System Design, Data Structures & Algorithms, OOP, Unit/Integration Testing, CI/CD, Git